Secure FSO Link Parts List

Part	Purpose
BPW34 photodiode	Optical receiver
D6505I laser diode	Optical transmitter
OPA320 op-amp	Amplifies photodiode current
SN74AHCT125 buffer	Drives the transmitter switching signal
LMV7219 comparator	Converts the received signal to digital levels
SOT-23-5 to DIP adapter	Mounts for ICs
2N7000 MOSFET	Switch for laser current
22 ohm resistor	Limits ringing and switching current
FIT0203 perfboard	For receiver and transmitter circuits
HK-6.4 laser housing	Housing for laser
25mm piano-convex lens	Focuses beam onto photodiode
Red acrylic color filter	Ambient light filter
Black pipe or shroud	Ambient light blocker
82 kΩ	OPA320 transimpedance feedback resistor
100 kΩ	Creates the receiver’s half-supply reference voltage
1 kΩ	Current limiting around the comparator and controls
100 Ω	Laser-driver/current-path resistance
1 MΩ	Bias or discharge path wherever will be needed
10 kΩ potentiometer	Adjustable comparator threshold
100 nF ceramic	One beside every IC supply pin/reference-voltage filtering
10 µF electrolytic	Bulk power filtering on each board
10 pF ceramic	OPA320 feedback compensation
22 pF ceramic	Alternative OPA320 feedback compensation
47 pF ceramic	Alternative OPA320 feedback compensation